ITCS216: Fundamentals of Database Systems (1/2026)

Faculty of Information and Communication Technology

In-class Exercise 09: Database Implementation

Objectives:

- Design a complete data dictionary from the given relational data model.
- Implement the relational model as a database using Data Definition Language (DDL).
- Insert and modify sample records using Data Manipulation Language (DML).

Instructions:

1. Given the following relational data model, complete the tasks below.
2. Submit the Data Dictionary in PDF and the *executable* `.sql` file containing the DDL and DML statements, named each file, "ITCS216-InclassActivity-Lect09-XX88XXX"

Part I: Complete the Data Dictionary

Using the given relational data model, business rules, and sample data, complete all missing information in the provided data dictionary.

TutorMe is an online tutoring platform that connects students with tutors. Each tutor teaches one main subject and specifies an hourly rate. Students can book tutoring sessions with available tutors.

Tutor	tutor_id	tutor_name	subject	email	hourly_rate	
TutoringSession	session_id	student_id	tutor_id	session_date	duration_min	session_status
Student	student_id	student_name	email			

- Each tutor is uniquely identified by `tutor_id`. A tutor must have a name, unique email address, subject, and positive hourly rate.
  Sample tutors:
  ```
  (101, 'Narin Chai', 'narin@tutorme.com', 'Mathematics', 450.00)
  (102, 'Bella Wong', 'bella@tutorme.com', 'English', 500.00)
  (103, 'Clara Kim', 'clara@tutorme.com', 'Programming', 550.00)
  ```
- Each student is uniquely identified by `student_id`. A student must have a name and unique email address.
  Sample students:
  ```
  (201, 'Alex Wen, 'alex@email.com')
  (202, 'Mina Lee', 'mina@email.com')
  (203, 'Peter Park', 'peter@email.com')
  ```
- Each tutoring session is uniquely identified by session_id.
  Sample tutoring sessions:
  ```
  (401, 201, 101, '2026-10-06', 60, 'Completed')
  (402, 202, 103, '2026-10-07', 90, 'Confirmed')
  (403, 203, 102, '2026-10-08', 60, 'Pending')
  (404, 201, NULL, '2026-10-10', 120, 'Pending')
  ```
- The session date must use the YYYY–MM–DD format.
- The session duration must be between 30 and 180 minutes.
- The session status must be “Pending”, “Confirmed”, “Completed”, or “Cancelled”.

Table Tutors

Attribute	Description	Data type	Format	Nullable	Range/Domain	Key	Referenced attribute
tutor_id	Unique identifier for a tutor						
tutor_name	Tutor’s full name						
subject	Main subject taught by the tutor						
email	Tutor’s email address						
hourly_rate	Tutoring fee per hour						

Table TutoringSession

Attribute	Description	Data type	Format	Nullable	Range/Domain	Key	Referenced attribute
session_id	Unique identifier for a tutoring session						
student_id	Identifies the student who books the session						
tutor_id	Identifies the tutor who conducts the session						
session_date	Scheduled date of the tutoring session						
duration_min	Duration of the tutoring session in minutes						
session_status	Status of the tutoring session						

Table Student

Attribute	Description	Data type	Format	Nullable	Range/Domain	Key	Referenced attribute
student_id	Unique identifier for a student						
student_name	Student's full name						
email	Student's email address						

Part II Create the DDL and DML Script

Create one executable SQL script that performs the following tasks:

1. Create a database named `TutorMeDB`.
2. Set `TutorMeDB` as the current database.
3. Create all tables in the correct sequence based on the data dictionary completed in Part I.
4. Define all primary keys, foreign keys, and other constraints specified in the Part I data dictionary.
5. Insert the provided sample data, with at least three records in each table.
6. Execute the complete script and correct any errors.
